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PROJECT PROPOSAL

SECTION 01B

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INTRODUCTION

Current Issues

The education sector continues to face significant challenges in managing teaching and learning activities effectively, particularly in institutions that lack a centralized and automated system for educational management. EduQuest, as a teaching and learning management system, addresses the critical operational difficulties that educational institutions encounter daily. Managing user roles and permissions remains a substantial challenge, as administrators, instructors, and students require different levels of access to educational resources and functionalities. This often results in confusion, security vulnerabilities, and inefficient resource allocation that hampers the learning process.

Course management presents another critical issue for educational institutions. Without a systematic approach, educators struggle to organize course materials, assignments, and assessments effectively. Manual updates and the absence of real-time synchronization lead to outdated or inconsistent course content, which directly impacts the quality of education delivered. Students often find themselves confused about assignment deadlines, course schedules, and learning materials, resulting in poor academic performance and decreased engagement.

The lack of an organized system to manage student enrollment and progress tracking creates a major operational bottleneck for educational institutions. Many institutions still rely on manual processes or fragmented systems for handling course registrations, attendance monitoring, and grade management. This inefficiency leads to data inconsistencies, administrative errors, and significant delays in providing students with timely feedback on their academic performance. The absence of integrated communication tools further exacerbates these problems, making it difficult for instructors and students to collaborate effectively outside the traditional classroom setting.

Assessment and assignment management pose additional challenges in the current educational landscape. Without a centralized platform, instructors must juggle multiple tools and platforms to create, distribute, and grade assignments. This fragmentation not only increases their workload but also makes it difficult to maintain consistent grading standards and provide comprehensive feedback. Students, on the other hand, struggle to keep track of

multiple submissions across different platforms, leading to missed deadlines and incomplete work.

Another significant issue is the absence of a systematic approach to monitoring student progress and generating meaningful analytics. Educational institutions often lack the tools necessary to identify at-risk students early, track learning outcomes effectively, or measure the success of their teaching methodologies. This deficiency prevents educators from making data-driven decisions to improve their teaching strategies and support students who need additional assistance.

These challenges underscore the urgent need for an integrated teaching and learning management system like EduQuest that can streamline educational operations, enhance communication, and improve learning outcomes. By implementing a centralized solution, educational institutions can automate course management, facilitate seamless assignment submissions and grading, track student progress in real-time, and leverage analytics to drive continuous improvement in teaching and learning practices. Such a system would not only reduce administrative burden and minimize errors but also create a more engaging and effective learning environment, ensuring better educational outcomes and preparing students for success in an increasingly digital world.

Objective of Study

The main objective of this study is to develop a comprehensive web-based Teaching and Learning Management System called EduQuest that enhances the efficiency of educational operations and improves the overall teaching and learning experience, with a particular focus on supporting small to medium-sized educational institutions, including those in rural or underserved areas where access to advanced educational technology is often limited. Specifically, the study aims to:

1. Develop a robust course management module that enables instructors to create, organize, and update course content efficiently, ensuring students have access to current and relevant learning materials at all times.
2. Implement an automated assignment and assessment system that streamlines the process of creating, distributing, submitting, and grading assignments, reducing administrative workload and providing timely feedback to students.
3. Create a student enrollment and progress tracking system that automates registration processes, monitors attendance, records grades, and generates comprehensive progress reports to help educators identify students who need additional support.
4. Integrate a communication and collaboration platform that facilitates seamless interaction between instructors and students through announcements, discussion forums, and messaging features, fostering an engaging learning community.
5. Develop an analytics and reporting dashboard that provides actionable insights into student performance, course effectiveness, and learning outcomes, enabling data-driven decision-making to continuously improve educational quality.

By addressing these objectives, the proposed EduQuest Teaching and Learning Management System will not only streamline operations for educational institutions but also bridge the technological gap for those in rural or underserved areas, empowering them to deliver high-quality education and compete effectively in the modern educational landscape. This system will ultimately contribute to improved learning outcomes, enhanced student engagement, and greater operational efficiency across all levels of education.

Project to be Develop

The project to be developed is EduQuest, a comprehensive Teaching and Learning Management System designed to address the current challenges faced by educational institutions in managing their academic operations efficiently. The system will be divided into three key modules to ensure comprehensive functionality and seamless educational management:

1. Module 1: Manage Profile

This module will handle user roles and permissions across different stakeholder groups within the educational institution. Students will be able to register and login to the system, view and update their personal information, academic details, and profile settings. Instructors will have the capability to create and manage their professional profiles, update contact information, and maintain their teaching credentials. Administrators will be granted comprehensive access to view, create, update, and delete user accounts when necessary, assign appropriate roles and permissions, and monitor user activity across the platform. The module will also enable users to upload profile pictures, manage account security settings, and maintain their personal academic portfolios. This ensures secure access control, streamlines user management processes, and provides a personalized experience for all users within the EduQuest system.

2. Module 2: Manage Assignments and Notes

This module will serve as the central hub for all academic content management, enabling instructors to create, upload, update, and delete course assignments with detailed instructions, deadlines, and grading rubrics. Instructors can organize and categorize learning materials, upload lecture notes in various formats (PDF, documents, presentations), and structure content by topics or weeks. Students will have the ability to access all course materials, download lecture notes for offline study, view assignment details and requirements, submit their completed assignments through the platform, and track submission status and grades. The system will support multiple file formats for both notes and assignment submissions, implement deadline reminders and notifications, and maintain a comprehensive archive of all academic materials. Administrators can monitor content quality, ensure compliance with institutional standards, and generate reports on assignment completion rates. This module

ensures an organized, efficient, and accessible academic content management system that enhances the teaching and learning experience for both instructors and students.

3. Module 3: Manage Announcement and Discussion

This module will serve as an integrated communication and collaboration platform within the EduQuest system, promoting active engagement and continuous interaction among administrators, instructors and students. It centralizes both announcement and discussion features to ensure that important academic information and discussions are easily accessible within one unified system. Instructors can create, edit and delete announcements to share course updates, reminders or academic notices, while students can view these announcements and participate in topic-based academic discussions. The discussion feature allows both instructors and students to post, reply and update discussion threads, as well as search or filter topics for quick access to relevant information. Administrators play a supervisory role by monitoring announcements and discussions to ensure appropriate communication and remove any unsuitable content when necessary. By integrating these communication tools, the module fosters a collaborative learning environment that enhances transparency, encourages knowledge sharing, and ensures timely dissemination of information, thereby strengthening engagement and connection between all users within the EduQuest platform.

By incorporating these three essential modules, EduQuest will effectively address the current inefficiencies faced by educational institutions, providing a centralized, automated, and user-friendly solution that streamlines academic operations, enhances communication between all stakeholders, and ultimately improves the quality of teaching and learning. This comprehensive system will empower educational institutions to deliver better educational outcomes while reducing administrative burden, helping them thrive in an increasingly competitive and technology-driven educational landscape.

PROBLEM / MOTIVATION

Educational institutions face significant challenges in managing their teaching and learning activities due to reliance on fragmented systems and manual processes. The absence of an integrated management system leads to inefficiencies in course content distribution, assignment management, communication breakdowns between instructors and students, and difficulties in tracking academic progress. Students often encounter outdated learning materials, unclear assignment instructions, missed announcements, or inconsistent access to course resources, severely affecting their learning experience and academic performance. Additionally, instructors struggle to organize course content effectively, manage multiple assignment submissions across different platforms, provide timely feedback, and maintain consistent communication with their students, leading to increased workload and reduced teaching effectiveness.

The lack of a centralized platform creates substantial operational bottlenecks for educational institutions. Administrators find it challenging to monitor academic activities, generate meaningful reports on student performance, and ensure standardized processes across different courses and departments. Students face difficulties in keeping track of assignment deadlines, accessing lecture notes, participating in class discussions, and staying informed about important announcements. This disorganization results in missed deadlines, incomplete coursework, reduced student engagement, and ultimately, poor learning outcomes. Furthermore, the absence of a systematic discussion and collaboration platform limits peer-to-peer learning opportunities and hinders the development of an engaged academic community.

To address these critical issues, EduQuest, a comprehensive Teaching and Learning Management System, is essential to streamline educational operations, ensuring efficient course content management, organized assignment workflows, seamless communication channels, and improved overall learning outcomes for both students and instructors. The system will eliminate the confusion caused by scattered resources, reduce administrative burden on educators, and create a more engaging and supportive learning environment.

With the increasing demand for digital transformation in education, particularly accelerated by the shift towards blended and online learning, EduQuest can bridge the technological gap by automating and streamlining the entire teaching and learning

management process. By offering a centralized platform for educational institutions, this system will significantly improve workflow efficiency, reduce operational errors, enhance student engagement and satisfaction, facilitate better instructor-student communication, and provide data-driven insights for continuous improvement in teaching methodologies and learning outcomes. EduQuest will empower educational institutions to deliver high-quality education in an organized, accessible, and technology-driven manner, ultimately preparing students for success in the digital age.

BACKGROUND OF PROJECT

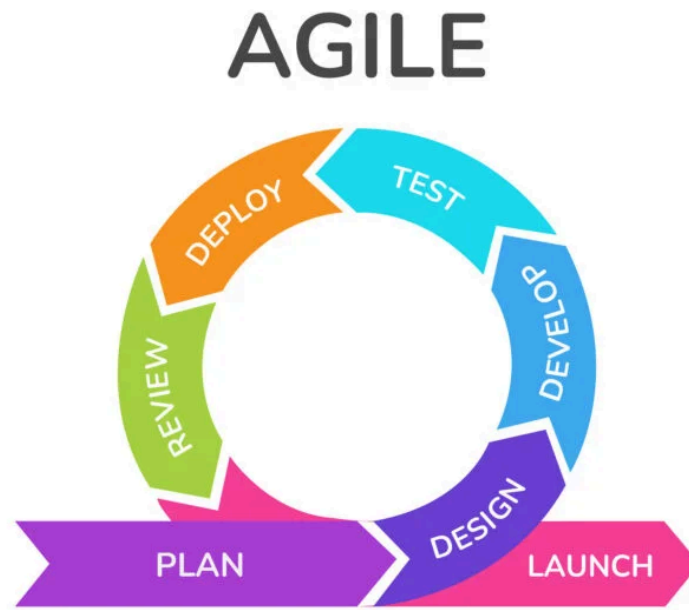
Most universities and colleges are moving towards digital learning because it is more flexible and efficient for both students and lecturers. However, not every institution has access to a proper management system that can handle all aspects of teaching and learning in one place. Many still rely on separate tools for sharing notes, giving assignments, and tracking attendance, which can be confusing and time-consuming.

The idea of developing a Teaching & Learning Management System (TLMS) came from observing how difficult it is to manage all these tasks manually or through different platforms. For example, some lecturers use Google Drive for notes, WhatsApp for announcements, and email for assignments. This scattered system often causes missed deadlines and miscommunication.

By using WordPress as the main platform, our group aims to build a TLMS that is simpler, easy to manage, and suitable for educational use. WordPress provides a user-friendly interface and many ready-made plugins that can support e-learning features like course creation, user management, and progress tracking. At the same time, it allows customization to suit the needs of lecturers and students.

This project will not only help improve the learning experience but also give us hands-on practice in software quality assurance, system design, and project development. We will be able to test, evaluate, and improve the system as part of our coursework, which aligns with the goals of this subject.

DEVELOPMENT METHODOLOGY



The development of the EduQuest system will apply the Agile methodology to ensure a flexible and efficient development process that emphasizes collaboration, adaptability, and iterative improvement. The proposed development process will be organized into several sprints, with each sprint focusing on a specific module such as Manage Profile, Manage Assignment and Notes, and Manage Announcement and Discussion. During each sprint, activities such as planning, design, coding, testing, and review will be conducted iteratively to refine functionality and system quality before progressing to the next phase. This approach will enable continuous feedback, early detection of potential issues and quick adaptation to changing user or system requirements. As a result, the final product is expected to meet the needs of both instructors and students effectively while maintaining high standards of usability, reliability and performance.

This methodology is selected because it supports adaptive planning and encourages frequent feedback from users, which is essential in educational system development where requirements may evolve based on instructor and student needs. Agile will allow the development team to respond quickly to changes, enhance user satisfaction through continuous improvement and ensure that the final system is well-aligned with user expectations and institutional objectives.

MODULE OF THE DEVELOPMENT

Module 1 : Manage Profile (Muhammad Haiqal Zafril Bin Zaidy CB22148)

Main users : Administrator, Instructor and Student

This module manages user roles, permissions and personal profiles within the EduQuest system. Students can register and log in, view and update personal information, academic details and profile settings. Instructors can create and maintain professional profiles, update contact information and manage teaching credentials. Administrators have full access to view, create, update, and delete user accounts, assign roles and permissions and monitor user activity. Users can also upload profile pictures, manage account security settings, and maintain personal academic portfolios.

Functions :

- 1) User registration and login system.
- 2) Edit and update personal profile personal and academic profile information.
- 3) Change or reset password.
- 4) Upload or update profile image.
- 5) Role-based profile view for administrator, instructor and student.
- 6) Administrators can assign roles, permissions, and monitor user activity.

CRUD Functions :

- 1) Create: User registers a new account with personal information.
- 2) Read: View profile details such as name, email, role and and permissions.
- 3) Update: Edit profile information such as name, email and password.
- 4) Delete: Users or administrators can permanently remove profiles from the system.

Expected outcome : A secure, efficient, and user-friendly profile management system that enables all stakeholders to manage personal and professional information accurately, maintain data privacy, and experience personalized access and functionality within EduQuest.

Module 2 : Manage Assignment and Notes (Qurratul Ain Damia Binti Muhammad Amran CB23125)

Main users : Administrator, Instructor and Student

This module enables instructors, students and administrators to efficiently manage academic materials and assignments within the EduQuest system. Instructors can create, upload, update and delete lecture notes and assignments, organize course content by topics or weeks and provide detailed instructions such as deadlines and grading criteria. Students can view and download learning materials and submit their completed work through the system. Administrators play a supervisory role by monitoring content quality, ensuring that uploaded materials meet institutional standards and overseeing assignment submissions for accuracy and completeness. The module applies CRUD (Create, Read, Update, Delete) operations to manage lecture notes and assignments systematically, allowing instructors and administrators to maintain an organized repository while enabling students to access resources and submit work efficiently.

Functions :

- 1) Upload and organize course notes and learning materials.
- 2) Create and manage assignments.
- 3) File upload for student assignment submission.
- 4) View and download uploaded materials.
- 5) Monitor and review content quality (admin role).

CRUD Functions :

- 1) Create: Instructor uploads new lecture notes or creates a new assignment with details and due dates.
- 2) Read: Students view uploaded notes and assignment details.
- 3) Update: Instructor edits assignment content or updates lecture materials.
- 4) Delete: Instructor removes outdated notes or completed assignments to keep the system organized and admin may delete invalid or inappropriate content.

Expected outcome : A centralized and organized academic management system that streamlines material sharing, assignment handling and content monitoring, ensuring

accessibility, accuracy and collaboration among instructors, students, and administrators within the EduQuest platform.

Module 3 : Manage Announcement and Discussion (Nur Alya Syakirah Binti Nasarudin CB22141)

Main users : Administrator, Instructor and Student

This module enhances communication and collaboration among administrators, instructors and students through integrated announcement and discussion features within the EduQuest system. Instructors can post important updates or course-related information, while students can participate in interactive academic discussions to exchange knowledge and ideas. Administrators play a supervisory role by monitoring all announcements and discussions, ensuring appropriate communication and maintaining content quality across the platform. The module applies CRUD (Create, Read, Update, Delete) operations to manage announcements and discussion topics efficiently, allowing for systematic organization and easy access to information within the system.

Functions :

- 1) Post and view announcements
- 2) Create and participate in discussion threads.
- 3) Comment or reply on discussions.
- 4) Edit or delete own announcements and discussion posts.
- 5) Search or filter announcements and topics.
- 6) Admin monitors, manages and removes inappropriate content.

CRUD Functions :

- 1) Create: Instructor posts new announcements and users (instructors and students) can create discussion topics.
- 2) Read: Users view all announcements and discussion threads.
- 3) Update: Instructor edits announcements and users can update their discussion posts.
- 4) Delete: Instructor, post owner or admin removes outdated announcements or inappropriate discussions.

Expected outcome : An interactive and collaborative platform that strengthens communication between administrators, instructors and students through timely announcements and academic discussions. It ensures proper content moderation,

keeps users informed and engaged, and fosters a supportive and well-organized learning environment within the EduQuest system.

EXPECTED OUTCOME

Our group expects to produce a fully working Teaching & Learning Management System (TLMS) that can be used by lecturers and students to manage their learning activities online. The system will make it easier to organize classes, share materials, and communicate without depending on multiple apps.

From the user side, instructors will be able to create courses, upload notes, give assignments, and check student submissions in one platform. Students will be able to view available courses, submit their work, check their marks, and communicate with their lecturers easily. The admin can manage users and monitor system activities through a simple dashboard.

The system will also include CRUD functions (Create, Read, Update, Delete) for courses, users, and learning materials to ensure data can be updated easily. Besides that, the project will focus on having a clean and simple interface that works well on both computer and mobile screens.

Overall, the expected outcome is a working prototype that shows how an online learning management system can improve teaching efficiency and student engagement. It will also give our team experience in handling real-world software development tasks such as planning, coding, testing, and maintaining the quality of a system.

CONCLUSION

In conclusion, the Teaching & Learning Management System (TLMS) project aims to provide a centralized and efficient platform to support teaching and learning activities for both lecturers and students. With the rapid shift towards digital education, especially after the increase in online and blended learning methods, it is important to have a system that can simplify how classes, assignments, and communication are managed.

Through this project, our group hopes to develop a system that is not only functional but also user-friendly and suitable for real academic use. By building it on WordPress, we are able to take advantage of its flexibility and ease of customization, while also learning how to apply the principles of Software Quality Assurance throughout the development process. This includes proper testing, validation, and documentation of each module to ensure the system runs smoothly and meets user expectations.

The TLMS will also serve as a learning experience for our team in terms of teamwork, project management, and problem-solving. Each member is responsible for a specific module, and by working together, we will be able to integrate all components into a complete and reliable system. At the same time, this project helps us understand the importance of maintaining software quality and ensuring that every feature works as intended.

Overall, the project represents our effort to contribute to digital learning innovation and to gain practical experience in developing real-world systems. Once completed, the Teaching & Learning Management System will not only demonstrate our technical and analytical skills but also provide a valuable tool that can enhance the learning experience for students and educators in any academic environment.

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